

Notes: Gathergood & Olafsson (RFS, 2026)

The Coholding Puzzle: New Evidence from Transaction-Level Data

Contents

1	Big picture	3
2	Incremental contribution	3
3	Data and measurement	3
3.1	Meniga financial aggregator data	3
3.2	Definition of coholding	3
3.3	Costs	4
3.4	Shocks and categories	4
4	Models and empirical specifications	4
4.1	Cross-sectional coholder regression	4
4.2	Shock-onset regression	4
4.3	Spending-category onset regression	4
5	Identification and empirical design	4
6	Tables and figures	5
6.1	Figure 1: Coholding deposit account balances and overdraft balances + Table 1: Coholding in the baseline sample	5
6.2	Figure 2: Share of days coholding + Table 2: Coholding summary statistics	5
6.3	Figure 3: Coholding spell duration vs. coholding frequency	6
6.4	Figure 4: Coholding by singles vs. multiperson households	6
6.5	Table 3: Comparison of coholders and non-co-holders	6
6.6	Table 4: Individual shocks and onset of coholding	6
6.7	Table 5: Excess Transactions on Overdraft Accounts	8
6.8	Table 6: OLS regression: Probability of cohoid period starting	8
6.9	Table 7: Individual fixed effects regression: Probability of cohoid period starting	10
7	Detailed result map	10
8	Short summary	11
9	Conclusion	11
9.1	Core conclusion	11
9.2	Incremental contribution	11
9.3	Economic intuition	11
9.4	Policy implications	11

9.5	Limitations	11
9.6	Takeaway	11

1 Big picture

- **Research question.** Why do consumers hold low-yield liquid savings while simultaneously paying high interest on unsecured revolving credit?
- **Core puzzle.** Simple arbitrage says the consumer should use liquid savings to reduce expensive debt unless there is a liquidity, coordination, behavioral, or informational reason not to repay.
- **Unique setting.** The paper studies Icelandic overdrafts, where interest accrues immediately, deposit balances earn near-zero interest, and both accounts are fully liquid.
- **Data advantage.** Meniga financial aggregator data provide daily balances and transaction-level spending categories across accounts.
- **Main descriptive result.** Coholding occurs on about 15% of individual-day observations, but most coholding is modest relative to consumption and short-lived.
- **Mechanism result.** The paper finds little support for household coordination failures or short-term shocks as the main explanations. The strongest evidence points to mental accounting and category-specific account use.

2 Incremental contribution

- **From survey snapshots to daily balance sheets.** Earlier studies mostly use surveys or statement balances. This paper shows that daily timing changes the interpretation of coholding.
- **From credit-card debt to overdraft coholding.** The overdraft setting removes credit-card grace-period and reward confounds.
- **From levels to spells.** The paper measures duration, frequency, and costs of coholding spells.
- **Consumption-normalized measurement.** Coholding is scaled by average daily consumption, converting balances into days of spending.
- **New evidence against standard explanations.** Household coordination and shocks receive weak support.
- **New evidence for mental accounting.** Transaction categories are systematically assigned to overdraft versus surplus accounts.

3 Data and measurement

3.1 Meniga financial aggregator data

The data come from Meniga, a financial aggregation platform used by roughly 20% of the Icelandic adult population. The sample period covers September 1, 2014 to January 31, 2017. The data contain daily account balances, transaction dates, amounts, merchant categories, account identifiers, and income arrivals.

Detailed interpretation. The platform creates a multi-account daily panel that allows the authors to see when coholding begins and ends.

3.2 Definition of coholding

$$\begin{aligned} Cohold_{it} &= 1\{Overdraft_{it} > 0, Deposits_{it} > 0\}. \\ CoholdLevel_{it} &= \min \left\{ \frac{OverdraftBalance_{it}}{AvgDailyConsumption_i}, \frac{DepositBalance_{it}}{AvgDailyConsumption_i} \right\}. \end{aligned}$$

Detailed interpretation. The minimum operator captures the part of debt that could be repaid using liquid deposits.

3.3 Costs

$$CoholdAmount_{it} = \min\{OverdraftBalance_{it}, DepositBalance_{it}\}, \quad DailyCost_{it} \approx CoholdAmount_{it} \times APR_i / 365.$$

Detailed interpretation. The cost is the interest saved if the consumer used coheld deposits to reduce the overdraft.

3.4 Shocks and categories

The paper constructs unemployment, health, and income shocks at monthly frequency and spending categories at daily frequency. The category analysis focuses on whether gambling, temptation, gasoline, alcohol, and other expenditures are disproportionately routed to overdraft accounts.

4 Models and empirical specifications

4.1 Cross-sectional cohorder regression

$$Coholder_i = \alpha + X_i' \beta + \varepsilon_i.$$

Detailed interpretation. This asks whether standard observables explain who ever coholds.

4.2 Shock-onset regression

$$StartCohold_{im} = \alpha + \beta_1 UnemploymentShock_{im} + \beta_2 HealthShock_{im} + \beta_3 IncomeShock_{im} + \varepsilon_{im}.$$

Detailed interpretation. This tests whether adverse shocks trigger the start of coholding spells.

4.3 Spending-category onset regression

$$StartCohold_{it} = \alpha + \gamma CategorySpending_{it} + \delta Controls_{it} + \varepsilon_{it}.$$

Detailed interpretation. This links the onset of coholding to transaction categories. The fixed-effect version adds individual effects α_i .

5 Identification and empirical design

The paper is primarily a high-frequency descriptive and mechanism study. It identifies how coholding behaves across days, spells, account types, household status, shocks, and spending categories. Daily data are crucial because monthly data make coholding appear more persistent than it is. The clean overdraft setting helps because overdrafts accrue interest immediately and offer no credit-card-style rewards or grace-period benefits. The mental-accounting evidence is correlational but strengthened by individual fixed effects.

6 Tables and figures

6.1 Figure 1: Coholding deposit account balances and overdraft balances + Table 1: Coholding in the baseline sample

Read: Figure 1 shows the joint distribution of deposits and overdrafts in consumption-days. Table 1 summarizes the same distribution in consumption-days and monetary units.

Detailed interpretation. Most observations lie on one axis, but the interior cells reveal true coholding. The interior mass is often modest rather than extreme.

Economic message. Coholding exists, but daily magnitudes are often small.

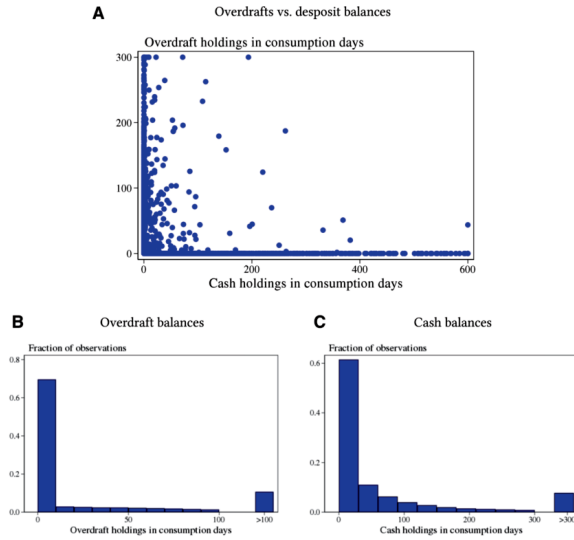


Figure 1
Coholding deposit account balances and overdraft balances
Note: that panel A shows a scatter plot of overdraft holdings and cash deposit account holdings, both measured in days of account-level average consumption expenditure. Panel B shows the distribution of overdraft holdings measured in days of account-level average consumption expenditure. Panel C shows the distribution of cash deposit account holdings measured in days of account-level average consumption expenditure. See Section 1 for details of sample restrictions.

Table 1
Coholding in the baseline sample

(A) Coholding in consumption days						
Overdraft holdings	Cash holdings					Total
	0	> 0–10	> 10–20	> 20–30	> 30	
0	2.95	15.65	6.79	4.88	35.47	65.73
> 0–10	1.65	1.27	0.19	0.10	0.63	3.83
> 10–20	1.23	1.12	0.15	0.07	0.35	2.93
> 20–30	1.16	1.04	0.14	0.07	0.28	2.70
> 30	11.50	9.90	1.02	0.55	1.83	24.81
Total	18.50	28.98	8.29	5.67	38.56	100.00

(B) Coholding in monetary units (000s ISK)						
Overdraft holdings	Cash holdings					
	0	> 0–20	> 20–40	> 40–60	> 60–80	> 80
0	2.95	8.82	3.76	2.92	2.40	44.87
> 0–20	0.60	0.38	0.05	0.03	0.03	0.35
> 20–40	0.45	0.30	0.04	0.03	0.02	0.23
> 40–60	0.41	0.31	0.05	0.02	0.03	0.20
> 60–80	0.39	0.26	0.04	0.02	0.02	0.17
> 80	13.70	9.57	1.22	0.79	0.57	29.84
Total	18.50	19.64	5.15	3.81	3.07	49.83

The table illustrates joint distribution of cash holdings (in deposit and/or savings accounts) and overdraft holdings in the baseline sample of individual $\times$ days. Panel A reports cash holdings and overdraft holdings normalized by average daily consumption spend of the consumer over the sample period. Panel B shows cash holdings and overdraft holdings in local currency. Each cell reports as percentage of observations. The cell (0,0) contains observations for which both cash and overdraft balance are zero.

(a) Figure 1: Coholding deposit account balances and overdraft balances

(b) Table 1: Coholding in the baseline sample

6.2 Figure 2: Share of days coholding + Table 2: Coholding summary statistics

Read: Figure 2 shows heterogeneity in the fraction of days with coholding. Table 2 summarizes frequency, duration, and cost.

Detailed interpretation. Many coholders cohold rarely, spells are short at the median, and costs are right-skewed.

Economic message. The welfare cost is concentrated in a right tail.

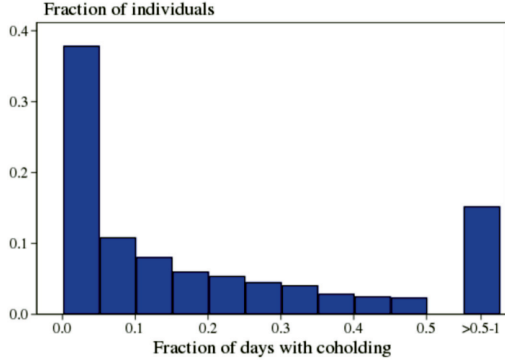


Figure 2
Share of days coholding
Note: that the figure shows the distribution of share of days that are coholding days, for the sample of accounts with at least one coholding day. Rightmost bin includes accounts for which the share is greater than 0.5.

(a) Figure 2: Share of days coholding

Table 2
Coholding summary statistics

(A) Share of days with coholding						
	Mean	SD	p50	p75	p90	p95
Min(3,3)	0.216	0.255	0.109	0.325	0.636	0.808
Min(5,5)	0.196	0.242	0.090	0.283	0.588	0.765
Min(10,10)	0.165	0.220	0.064	0.226	0.492	0.699
Min(15,15)	0.146	0.207	0.051	0.195	0.441	0.637
Min(20,20)	0.134	0.197	0.043	0.176	0.419	0.596
Min(25,25)	0.124	0.190	0.037	0.155	0.394	0.566
Min(30,30)	0.117	0.182	0.036	0.143	0.369	0.514

(B) Duration of coholding spells						
Duration of coholding (#days)						
	Mean	SD	P50	P75	P90	P95
Individual x day level	22.5	52.5	9.0	23.0	43.0	85.0
Individual level	29.6	61.4	13.0	27.6	65.5	112.2

C. Cost of coholding						
	Mean	SD	P50	P75	P90	P95
Individual x day:						
Annualized daily costs	4,702	16,673	0	1,403	12,793	26,917
Individual level:						
Average annual costs	4,702	9,813	1,198	4,548	12,518	20,823

In panel A each row of the table reports summary statistics for the level of coholding in the baseline sample, where the level is defined at the minimum of number of days' consumption held in overdraft balances and savings balance. For example, the first row reports that 21.6% of observations in the baseline sample with nonzero coholding show at least 3 days' consumption coheld in savings and overdraft balanced. Panel B reports summary data for duration of coholding spells, where a spell is defined as $Min(3, 3)$, holding 3 days consumption in both cash and overdrafts. Panel C presents measures of the cost of coholding. Annualized daily costs refer to the cost of coholding on-the-day for each individual $\times$ day observation and then multiply by 365 to create a simple annualized measure. Average annual costs report average annual costs from observed periods of 365 days. See Section 2.2 for further details of the calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

(b) Table 2: Coholding summary statistics

6.3 Figure 3: Coholding spell duration vs. coholding frequency

Read: The figure shows a negative relationship between spell length and the number of coholding spells.

Detailed interpretation. Frequent coholders tend to have shorter spells, while some infrequent coholders have longer spells.

Economic message. Coholding is dynamic account management, not only a static balance-sheet position.

6.4 Figure 4: Coholding by singles vs. multiperson households

Read: The scatterplots compare singles, linked individuals, and couples.

Detailed interpretation. Similar distributions weaken the household-coordination explanation.

Economic message. Coholding is not mainly caused by spouses failing to coordinate.

6.5 Table 3: Comparison of coholders and non-co-holders

Read: Coholders and non-coholders are similar in age, gender, and income; observables explain little of coholder status.

Detailed interpretation. Coholding is not well explained by standard demographics.

Economic message. The mechanism likely lies in account use and transaction behavior.

6.6 Table 4: Individual shocks and onset of coholding

Read: The table tests unemployment, health, and income shocks.

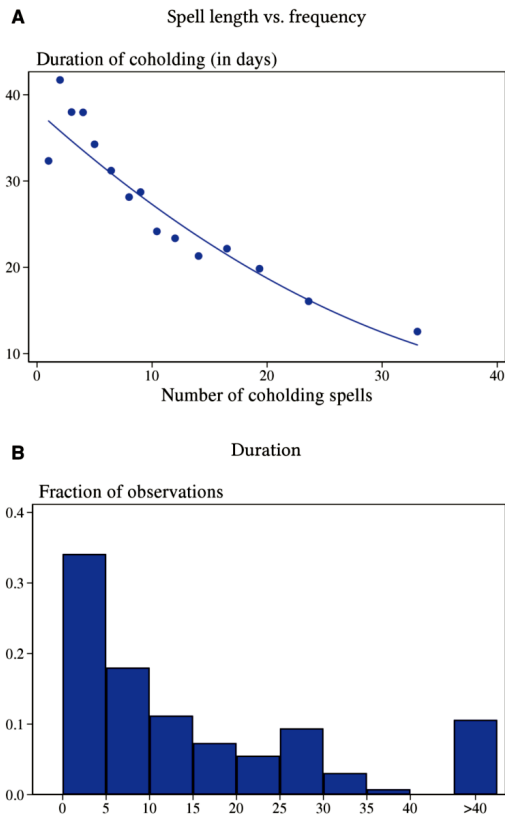


Figure 3: Coholding spell duration vs. coholding frequency

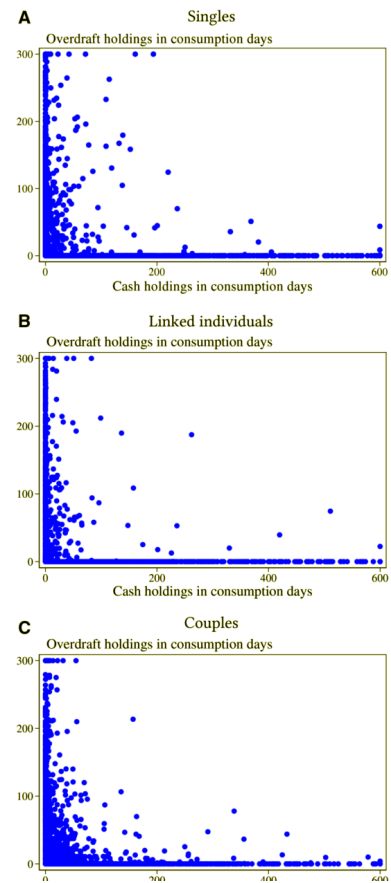


Figure 4: Coholding by singles vs. multiperson households

Table 3
Comparison of coholders and non-co-holders

(A) Summary statistics				
	Non-co-holder	Co-holder	t-value	p-value
Age (in years)	41.31	42.33	-4.25	0.000
Woman (=1)	0.49	0.48	1.35	0.171
Permanent Income (ISK/1,000,000)	0.87	0.94	-0.12	0.903
Couple / Linked (=1)	0.41	0.40	3.30	0.001
Social security (=1)	0.29	0.33	-3.81	0.000
(B) OLS regression				
	Co-holder = 1			
Age (in years)	0.000687*			
	(0.000373)			
Female (=1)	-0.00554			
	(0.00910)			
Permanent Income (ISK/1,000,000)	0.118***			
	(0.0164)			
Spouse (=1)	-0.0472***			
	(0.0132)			
Social Security (=1)	0.0392***			
	(0.00988)			
Constant	0.259***			
	(0.0167)			
R-squared	0.009			
#Observations	11,545			

Panel A presents a comparison of means between individuals who cohold (on at least one day of the sample period) and those who never cohold. Panel B draws the same sample and presents a cross-sectional regression in which the outcome variable is a 1/0 dummy indicating whether the individual is a coholder. * $p < .1$; ** $p < .05$;

Table 3: Comparison of coholders and non-co-holders

Table 4
Individual shocks and onset of coholding

(A) Summary statistics			
Variable	Obs	Mean	Std. dev.
Unemployment shock	111,580	0.007	0.082
Health shock	111,580	0.029	0.166
Income shock	111,580	0.101	0.301
(B) OLS regression			
	Probability of co-hold period starting		
	(1)	(2)	(3)
Unemployment shock	-0.0035 (0.0033)	0.0005 (0.0053)	-0.0792 (0.2074)
Health shock	-0.0011 (0.0022)	-0.0010 (0.0023)	0.0701 (0.1162)
Income shock	-0.0039*** (0.0012)	-0.0042*** (0.0012)	0.0507 (0.0626)
Unemployment shock_t-1		-0.0050 (0.0044)	
Health shock_t-1		0.0023 (0.0023)	
Income shock_t-1		0.0006 (0.0012)	
Unemployment shock_t-2		-0.0046 (0.0034)	
Health shock_t-2		0.0008 (0.0023)	
Income shock_t-2		0.0002 (0.0012)	
Unemployment # spread			0.0065 (0.0174)
Health # spread			-0.0059 (0.0096)
Income # spread			-0.0045 (0.0052)
Spread			0.0135*** (0.0017)
Constant	0.0146*** (0.0004)	0.0150*** (0.0005)	-0.1490*** (0.0209)
R-squared	.0001	.0002	.0007
#Observations	111,580	103,610	111,580

Panel A presents summary statistics for unemployment, health and income shocks in the sample of coholders. The unit of observation is an individual $\times$ month, with the shock variables in each cases coded to one if the individual experiences the shock in the month, and zero otherwise. See Section 3.3 for definitions of the shock

Table 4: Individual shocks and onset of coholding

Detailed interpretation. Shocks do not positively predict the start of coholding spells.

Economic message. Coholding does not primarily look like short-term financial distress.

6.7 Table 5: Excess Transactions on Overdraft Accounts

Read: Gambling, temptation, gasoline, and alcohol are disproportionately placed on overdraft accounts.

Detailed interpretation. The categories most routed to overdraft accounts are immediate or temptation-related.

Economic message. This is direct evidence of category-based mental accounting.

6.8 Table 6: OLS regression: Probability of cohold period starting

Read: The onset of coholding is positively related to spending categories such as gambling and temptation.

Detailed interpretation. The table links mental-accounting category patterns to the beginning of coholding spells.

Economic message. Coholding begins around particular transaction events.

Table 5
Excess Transactions on Overdraft Accounts

	Whole Sample	Age		Income	
		High	Low	High	Low
Gambling	1.18	1.13	1.24	1.14	1.24
Temptation	1.11	1.07	1.13	1.09	1.13
Gasoline	1.10	1.09	1.11	1.09	1.10
Alcohol	1.09	1.07	1.11	1.09	1.10
Pharmaceuticals	1.08	1.06	1.08	1.09	1.07
Transportation	1.07	1.08	1.05	1.06	1.07
Swim & leisure	1.06	1.09	1.03	1.07	1.04
Groceries	1.04	1.04	1.04	1.05	1.03
Special occasions	1.03	1.02	1.02	1.01	1.05
Online media	1.02	1.01	1.02	1.02	1.02
Online gaming	1.02	1.11	0.98	1.06	0.97
Toys	1.01	1.01	1.03	0.99	1.03
Ready made food	1.01	1.05	0.98	1.03	0.98
Clothes and accessories	1.01	1.01	0.99	1.02	0.99
Sports and activities	1.01	1.00	1.02	0.99	1.02
Home furnishings	1.00	0.99	1.00	1.01	1.00
Recreation	0.98	1.00	0.97	0.99	0.98
Hobbies	0.97	0.97	0.94	0.99	0.96
Cinema	0.97	1.06	0.91	1.02	0.92
Books & stationary	0.96	0.97	0.93	0.97	0.95
Home renovation	0.96	0.89	1.06	0.94	0.97
Holidays	0.94	0.93	0.93	0.93	0.95

The table presents summary data for the normalised share of transactions by category that are placed on an

Table 5: Excess Transactions on Overdraft Accounts

Table 6
OLS regression: Probability of cohort period starting

	Co-hold Spell Starting = 1				
	(1)	(2)	(3)	(4)	(5)
Age	0.0001*** (0.0000)	0.0000*** (0.0000)	0.0001*** (0.0000)	0.0001*** (0.0000)	0.0000*** (0.0000)
Woman	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	0.0000 (0.0001)
Linked	-0.0004*** (0.0002)	-0.0005** (0.0002)	-0.0003 (0.0002)	-0.0004*** (0.0002)	-0.0003 (0.0002)
Benefits person	0.0006*** (0.0001)	0.0006*** (0.0001)	0.0005*** (0.0001)	0.0006*** (0.0001)	0.0005*** (0.0001)
Log total income	0.0008*** (0.0000)	0.0006*** (0.0000)	0.0005*** (0.0000)	0.0005*** (0.0000)	0.0008*** (0.0001)
Payday ^a	-0.0043*** (0.0006)				-0.0049*** (0.0007)
Durables ^b		0.0001** (0.0000)			0.0000 (0.0000)
Non-durables ^b		0.0006*** (0.0001)			0.0001* (0.0001)
Nr. current accounts		0.0002** (0.0001)			0.0001 (0.0001)
Nr. savings accounts		0.0010*** (0.0000)			0.0010*** (0.0000)
Log cash spendings			0.0003*** (0.0000)		0.0003*** (0.0000)
Log credit card spendings				0.0001*** (0.0000)	0.0000*** (0.0000)
Gambling ^c					0.0014*** (0.0005)
Temptation ^c					0.0006** (0.0003)
Gasoline ^c					0.0005*** (0.0002)
Home Renovation ^c					0.0001 (0.0012)
Books & Stationary ^c					0.0002 (0.0005)
Holidays ^c					-0.0001 (0.0003)
Constant	0.0113*** (0.0004)	0.0112*** (0.0004)	0.0099*** (0.0004)	0.0111*** (0.0004)	0.0102*** (0.0004)
R-squared	0.002	0.003	0.002	0.002	0.003
#Observations	3,521,856	3,094,876	3,521,856	3,521,856	3,094,876

^aDummy that equals 1 if salary payment is made into the account on the day. ^bExpenditure on durable and nondurable goods in the preceding month expressed as a share of individual-specific average expenditures. ^cDummies that equal 1 if there is at least one expenditure on an item in that category on the day. Standard errors are clustered at the individual level. Individual location dummies are included in all models but coefficients not reported. Total income, cash and credit card balance are inverse-hyperbolic-sine transformed. Additional controls are day of week and day of month, but coefficients are not reported. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6: OLS regression: Probability of cohort period starting

6.9 Table 7: Individual fixed effects regression: Probability of cohoid period starting

Read: The fixed-effect estimates preserve the main category patterns.

Detailed interpretation. Within-person variation supports the transaction-category interpretation.

Economic message. Mental accounting is not purely cross-sectional sorting.

Table 7
Individual fixed effects regression: Probability of cohoid period starting

Co-hold period starting = 1

	(1)	(2)	(3)	(4)	(5)
Log total income	0.0008*** (0.0001)	0.0006*** (0.0000)	0.0005*** (0.0000)	0.0005*** (0.0000)	0.0008*** (0.0001)
Payday ^a	-0.0044*** (0.0010)				-0.0049*** (0.0012)
Durables ^b		0.0001*** (0.0000)			0.0000 (0.0000)
Non-durables ^b		0.0006*** (0.0001)			0.0001 (0.0001)
Nr. current accounts		0.0020*** (0.0006)			0.0016*** (0.0006)
Nr. savngs accounts		0.0024*** (0.0003)			0.0020*** (0.0003)
Log cash spendings			0.0003*** (0.0000)		0.0003*** (0.0000)
Log credit card spendings				0.0001*** (0.0000)	0.0000** (0.0000)
Gambling ^c					0.0013** (0.0005)
Temptation ^c					0.0007** (0.0003)
Gasoline ^c					0.0003 (0.0002)
Home Renovation ^c					0.0002 (0.0013)
Books & Stationary ^c					0.0002 (0.0005)
Holidays ^c					-0.0000 (0.0003)
Constant	0.0129** (0.0056)	0.0110 (0.0072)	0.0114* (0.0059)	0.0127** (0.0056)	0.0102 (0.0068)
R-squared	0.003	0.003	0.003	0.003	0.003
#Observations	3,522,740	3,095,136	3,522,740	3,522,740	3,095,136
#Individuals	3,985	3,985	3,985	3,985	3,985

^aDummy that equals 1 if salary payment is made into the account on the day. ^bExpenditure on durable and nondurable goods in the preceding month expressed as a share of individual-specific average expenditures. ^cDummies that equal 1 if there is at least one expenditure on an item in that category on the day. Standard errors are clustered at the individual level. Individual location dummies are included in all models but coefficients not reported. Total income, cash and credit card balance are inverse-hyperbolic-sine transformed. Additional controls

Table 7: Individual fixed effects regression

7 Detailed result map

- Daily cohoiding is common but not dominant.
- Cohoiding is typically modest relative to consumption.
- Monthly aggregation makes cohoiding look more persistent.
- Household coordination and shocks receive weak support.
- Category-level account routing provides the strongest mechanism evidence.

8 Short summary

The paper uses daily financial aggregator data to remeasure the coholding puzzle. It finds that coholding is often short-lived and modest, that standard observable characteristics and shocks explain little, and that spending categories predict account choice and coholding onset. The key mechanism is mental accounting across credit and debit accounts.

9 Conclusion

9.1 Core conclusion

Coholding is real, but high-frequency data show that many episodes are short, small, and category-specific rather than large persistent arbitrage failures.

9.2 Incremental contribution

The paper changes the measurement horizon and the mechanism. It shows that the household-finance puzzle cannot be understood from end-of-month balances alone; daily transaction flows reveal spell dynamics and category-based account use.

9.3 Economic intuition

Consumers appear willing to pay some interest costs to preserve mental budgets across accounts.

9.4 Policy implications

Disclosure of coholding costs may not be enough. Better interventions may need to preserve useful budgeting structure while reducing avoidable interest costs.

9.5 Limitations

The mental-accounting evidence is correlational, the setting is Icelandic overdrafts, and unlinked accounts could create measurement error.

9.6 Takeaway

Transaction-level data transform the coholding puzzle from a static balance-sheet anomaly into a dynamic account-choice phenomenon.